

Swimming in Chlorinated Pools: Evidence-Based Health Assessment

Executive Summary

Swimming in chlorinated pools involves exposure to chlorine disinfectants and their byproducts (DBPs), primarily chloramines. While chlorine effectively prevents waterborne pathogens and makes pools safer for recreational use, exposure carries both benefits and risks. Chloramines in the water irritate skin, eyes, and the respiratory tract (including the nose) when they turn into gas into the air above the water, particularly indoors. The health impact depends on exposure duration, concentration levels, individual sensitivity, and whether swimming occurs indoors or outdoors. Competitive swimmers are the most possible victims of DBPs exposure, because they have to do a lot of high intensive training in swimming pools. For most recreational swimmers, risks are minimal when pools maintain proper chemical balance, but vulnerable populations and occupational exposures require special consideration.

Definition & Context

Chlorinated swimming pools use chlorine-based disinfectants (typically sodium hypochlorite, calcium hypochlorite, or chlorine gas) to maintain water quality and prevent microbial contamination. Chlorine is now well-known to lead to the formation of disinfection byproducts (DBPs) when reacting with organic matter introduced by swimmers. Common settings include public pools, private residential pools, competitive training facilities, water parks, and spas. Since chlorine is the most widely used disinfectant, some level of chloramines is expected in indoor air at public swimming pools and spas. Exposure occurs through direct skin contact, inhalation of vapors (especially indoors), and inadvertent water ingestion during swimming activities.

Benefits / Positive Effects

Benefit	Why It Matters	Supporting Evidence
Pathogen Prevention	Prevents transmission of recreational water illnesses (RWIs) including Cryptosporidium, E. coli, and norovirus	Prevent swimmer's ear, Crypto, diarrhea, hot tub rash, and other swimming-related illnesses (CDC, 2025)
Water Quality Maintenance	Ensures microbiologically safe swimming environment for multiple users	Microbiological monitoring of water quality was introduced to ensure that water is not harmful to human health (ScienceDirect, 2011)
Public Health Protection	Enables safe mass recreational swimming and competitive sports	Chlorinating products for home swimming pools are generally safe for swimmers when used properly (Poison Control)
Cost-Effective Disinfection	Provides affordable, widely available water treatment method	Industry standard practice with established safety protocols

Risks / Negative Effects

Risk	Potential Impact	Evidence / Source
Respiratory Irritation	Mild to Moderate - Coughing, wheezing, asthma exacerbation	Chloramines in the water irritate skin, eyes, and the respiratory tract (CDC, 2025)
Skin & Eye Irritation	Mild - Redness, dryness, burning sensation	Chloramines in the water irritate skin, eyes, and the respiratory tract (CDC, 2025)
Allergic Sensitization	Moderate - Development of chemical sensitivities	This then causes changes in the airway and promotes the development of allergic sensitivities (Swimming World, 2021)
Occupational Exposure	Moderate to Severe - Long-term health effects for pool workers	Some CBPs may affect the respiratory and skin health of those who stay indoor for long periods, such as swimming instructors, pool staff (PubMed, 2021)
Infrastructure Corrosion	Environmental - Metal degradation in facilities	chloramines can also contribute to corrosion of metals around the aquatic venue and in air handling systems (CDC, 2025)

Toxicity & Safety Profile

Short-term exposure: Generally low toxicity for recreational swimmers. Immediate effects include mild skin/eye irritation and potential respiratory symptoms, particularly in indoor facilities with poor ventilation.

Long-term exposure: Health effects from swimming in chlorinated pools and the corresponding stress reactions occurring in our bodies are unclear. Competitive swimmers and pool workers face higher cumulative exposure risks.

Dose/Exposure thresholds: The EPA limit is 4 ppm, but that is based on drinking water standards, though commercial pools follow varying state regulations ranging from 3-10 ppm maximum levels.

Health agencies' position:

- **CDC:** Recommends proper pool maintenance and pre-swim hygiene to minimize chloramine formation
- **EPA:** Sets 4 ppm maximum for drinking water; regulates pool chemical safety through FIFRA labeling
- **WHO:** No specific swimming pool chlorine guidelines found in current search

Vulnerable groups:

- Children (more sensitive respiratory systems)
- Individuals with asthma or respiratory conditions
- Pregnant women (limited research on DBP exposure)
- Pool workers and competitive swimmers (occupational/high-volume exposure)

Environmental Impact

Air quality: Indoor pools can accumulate chloramine vapors, affecting air quality for swimmers and spectators. Swimmers, bathers and spectators at indoor public swimming pools may experience difficulty breathing or other health-related problems due to a buildup of chloramines in indoor air.

Water systems: Chlorinated pool water discharge may impact local water treatment systems and aquatic ecosystems if not properly managed.

Chemical handling: Hazards of pool water treatment and maintenance chemicals (e.g., chlorine), and the protective measures pool owners should take to prevent fires, toxic vapor releases, and injuries present environmental and safety concerns during storage and use.

Who Should / Should Not Be Exposed

Recommended for:

- General recreational swimmers with proper pool maintenance
- Individuals without respiratory sensitivities
- Those following pre-swim hygiene practices

Exercise caution:

- People with asthma or respiratory conditions
- Individuals with sensitive skin or eyes

- Pregnant women (consult healthcare provider)

Minimize exposure:

- Pool workers (use proper ventilation and PPE)
- Competitive swimmers (consider training schedule modifications)
- Young children (limit exposure duration, ensure proper ventilation)

Safer Alternatives or Mitigation Strategies

Pre-swim practices:

- Showering before using the pool or spa will greatly reduce the amount of chloramines in indoor air
- Avoid urinating in pools to reduce organic load

Facility improvements:

- Ensure adequate ventilation in indoor pools
- Maintain proper water chemistry balance
- Regular pool maintenance and monitoring

Alternative disinfection:

- UV light systems (supplementary)
- Ozone treatment (combined with reduced chlorine)
- Saltwater systems (still produce chlorine via electrolysis)
- Mineral systems (supplementary treatment)

Personal protection:

- Rinse immediately after swimming
- Use goggles to protect eyes
- Limit time in poorly ventilated indoor pools

Comparison

Option	Pros	Cons	Best For
Chlorinated Pools	Effective pathogen control, widely available, cost-effective	Respiratory/skin irritation, chemical odor, DBP formation	General recreational swimming, public facilities
Saltwater Pools	Gentler on skin/eyes, less chemical odor	Still produces chlorine, higher initial cost, equipment maintenance	Residential pools, sensitive swimmers
UV + Reduced Chlorine	Lower chemical exposure, effective disinfection	Higher upfront cost, requires backup disinfection	Indoor facilities, health-conscious facilities
Natural/Biological Pools	No chemical exposure, environmentally friendly	Limited pathogen control, higher maintenance, climate-dependent	Low-use residential, environmental enthusiasts

Sources

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10. Anne Arundel County Health Department - Indoor Air Quality and Public Swimming Pools:

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